TASK-2

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GRP 108
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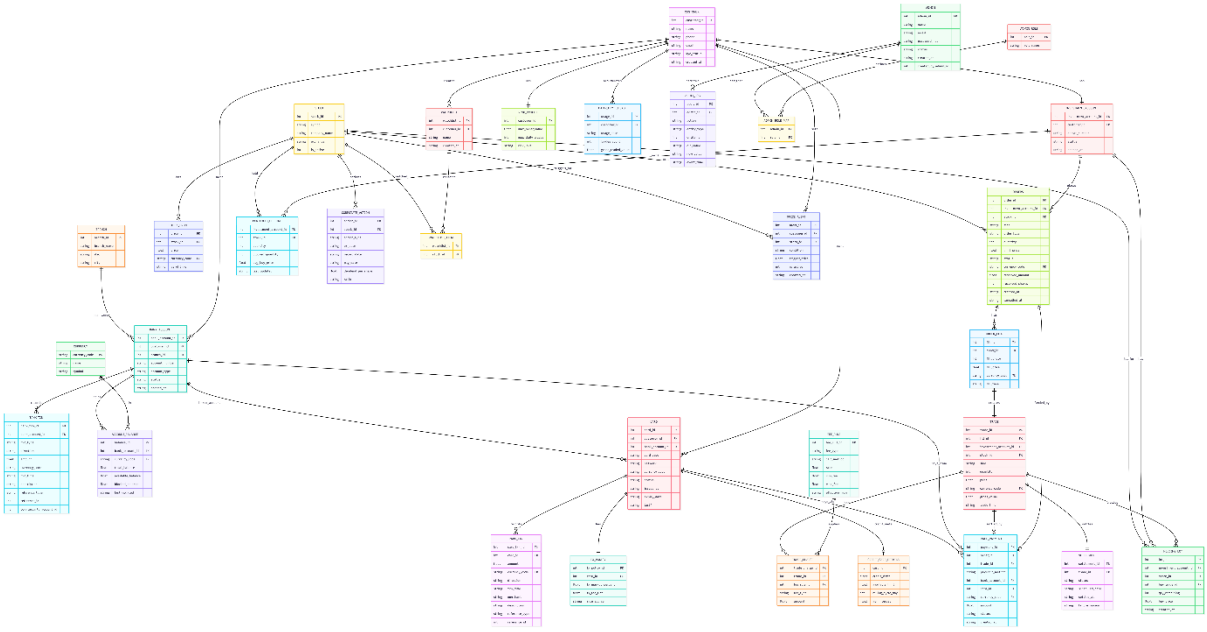
Relational model:



Drive link: https://drive.google.com/file/d/1GoJYhhQW6-T6Q3JTOefKBBI9cKASviM0/view?usp=sharing

Syntax used:

- ||--o{ = **one-to-many** (1 on left, 0..*many on right)
- ||--|| = **one-to-one** (exactly one on both sides)
- ||--o| = **one-to-zero-or-one** (optional 1 on right)

Weak entity sets:

- ADMIN_ROLE_MAP
- WATCHLIST_ITEM
- PORTFOLIO_HOLDING

Design Rationale

- The schema models real-world banking and trading operations in a structured manner.
- Core entities are kept independent to avoid redundancy and improve clarity.

Relationship Mapping

- One-to-many relationships are used where a single entity controls multiple dependent records.
- One-to-one relationships are enforced where exclusivity is required.
- Many-to-many relationships are resolved using associative relations.

Constraints and Integrity

- Primary and foreign keys enforce referential integrity.
- Unique constraints are applied to key business attributes.
- Centralized currency management ensures consistency across transactions

Normalization

- The schema follows normalization principles to minimize data duplication.
- Transactional and master data are stored separately.
- Historical data is maintained without affecting current records.

Scalability

- The design supports future feature expansion without major schema changes.
- Modular structure allows independent updates to system components.